

Mathematical Proof III: Constructing a Mathematical Proof¹

1 THE GOAL

In the first two sessions, we learned to identify what is given, determine what must be shown, unpack definitions, work forward and backward, search for intermediate results, and test examples.

Today we assume that we have found the basic idea.

The new question is:

I understand why the statement should be true.
How do I turn that idea into a rigorous proof?

A mathematical proof is not simply a sequence of calculations. It is a sequence of statements in which each step follows logically from something that came before it.

Our objective today is to practice turning

intuition $\longrightarrow$ proof skeleton $\longrightarrow$ complete argument.

2 A PROOF IS AN ARGUMENT

Suppose we want to establish

$$A \implies D.$$

¹Instructors: Yue Yu and Sofia Olguin. This note was prepared for the 2026 UCSB Math Camp for Ph.D. students in economics. It incorporates materials from previous instructors, including Shu-Chen Tsao, ChienHsun Lin, and Sarah Robinson.

During exploration, we may discover intermediate statements B and C .

A proof organizes them into a valid chain:

$$A \implies B \implies C \implies D.$$

But writing

$$A, \quad B, \quad C, \quad D$$

is not enough.

The reader needs to understand:

- why A implies B ;
- why B implies C ;
- why C implies D ;
- and why D is exactly the desired conclusion.

Compare the following two arguments.

Argument 1

$$(x - y)^2 \geq 0.$$

$$x^2 + y^2 \geq 2xy.$$

Argument 2

Since

$$(x - y)^2 \geq 0,$$

expanding gives

$$x^2 - 2xy + y^2 \geq 0.$$

Rearranging,

$$x^2 + y^2 \geq 2xy.$$

The mathematics is identical.

The second argument makes the logical connection between the steps explicit.

A proof should make clear not only what is true, but why the next step follows.

3 FROM A PROOF IDEA TO A PROOF SKELETON

Consider a familiar result.

Proposition 1

Let $C \subseteq \mathbb{R}^n$ be convex and let $f : C \rightarrow \mathbb{R}$ be strictly concave. Then f has at most one maximizer on C .

Suppose our exploration has already produced the following idea:

If there were two different maximizers, a point between them would be feasible. Strict concavity would make that point strictly better, which is impossible.

This is good mathematical intuition.

But it is not yet a proof.

The next step is to turn the intuition into a **proof skeleton**.

1. Suppose there are two distinct maximizers x^* and y^* .
2. Construct a point between them:

$$z = \lambda x^* + (1 - \lambda)y^*, \quad \lambda \in (0, 1).$$

3. Show that z is feasible.
4. Compare $f(z)$ with $f(x^*)$ and $f(y^*)$.
5. Obtain a value strictly larger than the maximum.
6. Conclude that two distinct maximizers cannot exist.

Notice what the skeleton does.

It identifies the *logical jobs* that must be completed without yet worrying about every algebraic detail.

4 FILLING IN THE SKELETON

Now we turn each step into a complete argument.

Step 1: Introduce the contradiction

Suppose, toward a contradiction, that there exist two distinct maximizers $x^*, y^* \in C$.

Since both are maximizers, there exists a maximal value M such that

$$f(x^*) = f(y^*) = M.$$

Step 2: Construct the useful object

Take any $\lambda \in (0, 1)$ and define

$$z = \lambda x^* + (1 - \lambda)y^*.$$

Step 3: Establish feasibility

Because C is convex and $x^*, y^* \in C$,

$$z \in C.$$

Step 4: Use strict concavity

Because $x^* \neq y^*$ and $\lambda \in (0, 1)$, strict concavity gives

$$\begin{aligned} f(z) &= f(\lambda x^* + (1 - \lambda)y^*) \\ &> \lambda f(x^*) + (1 - \lambda)f(y^*) \\ &= \lambda M + (1 - \lambda)M \\ &= M. \end{aligned}$$

Step 5: Identify the contradiction

But $z \in C$, and M is the maximum value of f on C .

Therefore

$$f(z) \leq M.$$

We have obtained both

$$f(z) > M \quad \text{and} \quad f(z) \leq M,$$

a contradiction.

Solution

Suppose, toward a contradiction, that $x^*, y^* \in C$ are two distinct maximizers. Let

$$f(x^*) = f(y^*) = M.$$

Take any $\lambda \in (0, 1)$ and define

$$z = \lambda x^* + (1 - \lambda)y^*.$$

Since C is convex, $z \in C$. Since f is strictly concave and $x^* \neq y^*$,

$$\begin{aligned} f(z) &> \lambda f(x^*) + (1 - \lambda)f(y^*) \\ &= M. \end{aligned}$$

Thus z is a feasible point satisfying $f(z) > M$, contradicting the fact that M is the maximum value of f on C .

Therefore f has at most one maximizer.

5 EVERY ASSUMPTION NEEDS A JOB

When constructing a proof, keep track of where each assumption is used.

In the previous proof:

$$C \text{ convex} \implies z \in C,$$

$$x^* \neq y^* \implies \text{strict concavity can be applied,}$$

$$f \text{ strictly concave} \implies f(z) > M,$$

$$x^*, y^* \text{ maximizers} \implies M \text{ is the largest feasible value.}$$

This gives a useful diagnostic question:

Where in my proof did I use each assumption?

If an important assumption never appears, one of three things may be happening:

- (1) the proof is incomplete;
- (2) the assumption is unnecessary;
- (3) the assumption was used implicitly and should be made explicit.

6 EXERCISE: ASSEMBLE THE PROOF

The following pieces belong to a proof of the proposition:

If $f : C \rightarrow \mathbb{R}$ is concave and x^*, y^* are maximizers of f on a convex set C , then every convex combination of x^* and y^* is also a maximizer.

The pieces are deliberately out of order.

- (A) Since M is the maximum value and $z \in C$,

$$f(z) \leq M.$$

- (B) Let

$$z = \lambda x^* + (1 - \lambda)y^*, \quad \lambda \in [0, 1].$$

- (C) Therefore

$$f(z) = M,$$

so z is a maximizer.

- (D) Since C is convex,

$$z \in C.$$

- (E) Since x^* and y^* are maximizers,

$$f(x^*) = f(y^*) = M.$$

(F) By concavity,

$$f(z) \geq \lambda f(x^*) + (1 - \lambda)f(y^*) = M.$$

Task 1

Put the pieces in a logically valid order.

Task 2

Identify the justification for each transition.

Task 3

Rewrite the pieces as a complete proof in your own words.

7 DO NOT HIDE THE KEY STEP

A common problem in proof writing is skipping exactly the step that matters.

Consider:

Proposition 1

If f and g are convex and $\alpha, \beta \geq 0$, then

$$h = \alpha f + \beta g$$

is convex.

A student might write:

Since f and g are convex,

$$h(\lambda x + (1 - \lambda)y) \leq \lambda h(x) + (1 - \lambda)h(y).$$

Therefore h is convex.

The conclusion is correct, but the main mathematical step has been hidden.

The proof should show how convexity of f and g produces convexity of h .

Solution

Take arbitrary x, y in the domain and $\lambda \in [0, 1]$. Then

$$\begin{aligned} h(\lambda x + (1 - \lambda)y) &= \alpha f(\lambda x + (1 - \lambda)y) + \beta g(\lambda x + (1 - \lambda)y) \\ &\leq \alpha [\lambda f(x) + (1 - \lambda)f(y)] + \beta [\lambda g(x) + (1 - \lambda)g(y)] \\ &= \lambda [\alpha f(x) + \beta g(x)] + (1 - \lambda) [\alpha f(y) + \beta g(y)] \\ &= \lambda h(x) + (1 - \lambda)h(y). \end{aligned}$$

Thus h is convex.

The second line is the key step.

It also reveals exactly why we need

$$\alpha, \beta \geq 0 :$$

multiplying an inequality by a nonnegative number preserves its direction.

8 EXERCISE: FILL IN THE MISSING JUSTIFICATIONS

Complete the following proof by explaining the justification for each marked step.

Proposition 1

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are increasing. Then

$$h(x) = f(x) + g(x)$$

is increasing.

Solution

Take $x, y \in \mathbb{R}$ with $x \leq y$.

Since f is increasing,

$$f(x) \leq f(y).$$

Justification: _____

Similarly,

$$g(x) \leq g(y).$$

Justification: _____

Adding the two inequalities,

$$f(x) + g(x) \leq f(y) + g(y).$$

Justification: _____

Therefore

$$h(x) \leq h(y).$$

Hence h is increasing.

Why does this final statement follow? _____

9 INTUITION IS NOT YET A PROOF

Consider the claim:

Proposition 1

If A is positive definite, then A is nonsingular.

An intuitive explanation might be:

A positive definite matrix cannot send a nonzero vector to zero, so it must be invertible.

This is correct intuition.

But the proof needs to explain *why* positive definiteness rules this out.

Solution

Suppose $Ax = 0$. Then

$$x^\top Ax = x^\top 0 = 0.$$

If $x \neq 0$, positive definiteness implies

$$x^\top Ax > 0,$$

which is impossible.

Therefore

$$Ax = 0 \implies x = 0.$$

Hence the null space of A is trivial, so A is nonsingular.

The proof translates the intuitive phrase

“cannot send a nonzero vector to zero”

into a sequence of mathematical implications.

10 EXERCISE: TURN INTUITION INTO PROOF

For each statement below, an intuitive argument is provided.

Your task is to turn the intuition into a complete proof.

10.1 Intersection of Convex Sets

Proposition 1

If A and B are convex, then $A \cap B$ is convex.

Intuition:

A point between two points that belong to both sets remains in both sets.

Your proof:

10.2 Strict Monotonicity

Proposition 2

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing, then f is injective.

Intuition:

Two different inputs cannot produce the same output because the larger input must produce a larger output.

Your proof:

10.3 Unique Utility Maximization

Proposition 3

Suppose $B \subseteq \mathbb{R}^n$ is convex and $u : B \rightarrow \mathbb{R}$ is strictly concave. If a maximizer exists, it is unique.

Intuition:

If there were two different optimal bundles, a bundle between them would give strictly higher utility.

Your proof:

11 MAKE THE DIRECTION OF THE ARGUMENT CLEAR

Consider a statement of the form

$$P \implies Q.$$

A proof should make clear that we begin with P and eventually establish Q .

For example:

Proposition 1

Suppose $x, y > 0$. If

$$x + \frac{1}{x} = y + \frac{1}{y}$$

and $x, y > 1$, then $x = y$.

One possible route is to define

$$f(t) = t + \frac{1}{t}.$$

For $t > 1$,

$$f'(t) = 1 - \frac{1}{t^2} > 0.$$

Thus f is strictly increasing on $(1, \infty)$, and hence injective there.

Therefore

$$f(x) = f(y) \implies x = y.$$

The important point is that the proof explicitly connects

$$x + \frac{1}{x} = y + \frac{1}{y}$$

to

$$f(x) = f(y),$$

then uses strict monotonicity to obtain the desired conclusion.

12 PROOFS WITH TWO DIRECTIONS

Some statements require more than one argument.

If the goal is

$$P \iff Q,$$

we need to prove both

$$P \implies Q$$

and

$$Q \implies P.$$

Similarly, if the goal is

$$A = B,$$

a common strategy is to prove

$$A \subseteq B \quad \text{and} \quad B \subseteq A.$$

Consider:

Proposition 1

For any function $f : X \rightarrow Y$ and sets $A, B \subseteq Y$,

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B).$$

A complete proof should establish both inclusions.

Solution

First, let

$$x \in f^{-1}(A \cap B).$$

Then

$$f(x) \in A \cap B,$$

so

$$f(x) \in A \quad \text{and} \quad f(x) \in B.$$

Hence

$$x \in f^{-1}(A) \quad \text{and} \quad x \in f^{-1}(B),$$

which implies

$$x \in f^{-1}(A) \cap f^{-1}(B).$$

Therefore

$$f^{-1}(A \cap B) \subseteq f^{-1}(A) \cap f^{-1}(B).$$

Conversely, let

$$x \in f^{-1}(A) \cap f^{-1}(B).$$

Then

$$f(x) \in A \quad \text{and} \quad f(x) \in B.$$

Thus

$$f(x) \in A \cap B,$$

which implies

$$x \in f^{-1}(A \cap B).$$

Therefore

$$f^{-1}(A) \cap f^{-1}(B) \subseteq f^{-1}(A \cap B).$$

Hence

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B).$$

13 EXERCISE: DID WE PROVE BOTH DIRECTIONS?

Suppose someone wants to prove

$$x^2 = y^2 \iff |x| = |y|.$$

They write:

If $x^2 = y^2$, then taking square roots gives

$$|x| = |y|.$$

Therefore

$$x^2 = y^2 \iff |x| = |y|.$$

Answer the following.

- (a) Which direction has been proved?
- (b) Which direction is missing?
- (c) Complete the proof.

14 CHOOSE THE RIGHT LEVEL OF DETAIL

A proof should be rigorous, but rigor does not mean explaining every piece of basic algebra.

Compare:

Too compressed

By concavity, the result follows.

This hides the main argument.

Too expanded

Since $\lambda \in [0, 1]$, we know $\lambda \geq 0$. Also, $1 - \lambda \geq 0$. Since $\lambda + (1 - \lambda) = 1$, the expression $\lambda x + (1 - \lambda)y$ is a convex combination...

This may be unnecessary if the audience already knows what a convex combination is.

Appropriate

Since C is convex,

$$\lambda x + (1 - \lambda)y \in C.$$

By concavity,

$$f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y).$$

The right level of detail depends on the audience, but the central logical steps should always be visible.

Do not explain every algebraic operation.
Do explain every important logical transition.

15 A FIRST-YEAR MICRO EXAMPLE: MONOTONIC TRANSFORMATIONS

Utility representations provide a useful example of proof construction.

Suppose $u : X \rightarrow \mathbb{R}$ represents a preference relation $\succeq$, meaning

$$x \succeq y \iff u(x) \geq u(y).$$

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be strictly increasing and define

$$v(x) = g(u(x)).$$

We want to show that v represents the same preference relation.

The proof idea is simple:

A strictly increasing transformation preserves the ranking of utility numbers.

But we need to translate this intuition into a formal argument.

Our goal is

$$x \succeq y \iff v(x) \geq v(y).$$

We already know

$$x \succeq y \iff u(x) \geq u(y).$$

Because g is strictly increasing,

$$u(x) \geq u(y) \iff g(u(x)) \geq g(u(y)).$$

Finally,

$$g(u(x)) = v(x) \quad \text{and} \quad g(u(y)) = v(y).$$

Putting the pieces together gives

$$x \succeq y \iff u(x) \geq u(y) \iff g(u(x)) \geq g(u(y)) \iff v(x) \geq v(y).$$

Solution

For any $x, y \in X$,

$$x \succeq y \iff u(x) \geq u(y),$$

because u represents $\succeq$.

Since g is strictly increasing,

$$u(x) \geq u(y) \iff g(u(x)) \geq g(u(y)).$$

By the definition $v = g \circ u$,

$$g(u(x)) = v(x) \quad \text{and} \quad g(u(y)) = v(y).$$

Therefore

$$x \succeq y \iff v(x) \geq v(y),$$

so v represents the same preference relation.

16 EXERCISE: BUILD THE ARGUMENT FROM INGREDIENTS

Suppose a consumer has budget set

$$B(p, w) = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\},$$

where $p \gg 0$ and $w > 0$.

Suppose

$$x^* \in \operatorname{argmax}_{x \in B(p, w)} u(x).$$

Assume u is strictly increasing in every component.

We want to show that

$$p \cdot x^* = w.$$

In words: an optimizing consumer with strictly monotone preferences exhausts the budget.

You are given the following proof idea:

If the consumer did not spend all wealth, there would be some money left. The consumer could use it to increase consumption of at least one good. That bundle would still be affordable and would give strictly higher utility, contradicting optimality.

Turn this idea into a rigorous proof.

1. What should you suppose if the conclusion fails?

2. If $p \cdot x^* < w$, define the slack

$$\varepsilon = \underline{\hspace{2cm}} > 0.$$

3. Construct a bundle with slightly more of good i

Let e_i denote the i -th standard basis vector.

Find some $\delta > 0$ such that

$$x' = x^* + \delta e_i$$

is still affordable.

4. What does strict monotonicity imply?

5. Where does the contradiction occur?

6. Write the complete proof.

17 PROOF SKELETONS AS A HABIT

Before writing paragraphs, try reducing your argument to a few lines.

For example:

$$\begin{array}{c} p \cdot x^* < w \\ \Downarrow \\ \text{construct } x' > x^* \text{ with } p \cdot x' \leq w \\ \Downarrow \\ u(x') > u(x^*) \\ \Downarrow \\ \text{contradiction to optimality} \end{array}$$

Once this skeleton is correct, writing the formal proof becomes much easier.

The skeleton also makes it easier to detect missing steps.

For example, the statement

$$p \cdot x^* < w \implies \text{there exists an affordable } x' > x^*$$

needs justification.

That is where the assumptions

$$p \gg 0 \quad \text{and} \quad w - p \cdot x^* > 0$$

enter the argument.

18 EXERCISE: SKELETON FIRST, PROOF SECOND

For each proposition:

1. write a proof skeleton using no more than five lines;
2. identify which assumption justifies each line;
3. only then write the complete proof.

18.1 Convex Combination of Convex Functions

Let $f, g : C \rightarrow \mathbb{R}$ be convex and let

$$\alpha, \beta \geq 0.$$

Show that

$$h = \alpha f + \beta g$$

is convex.

Proof skeleton:

Complete proof:

18.2 Positive Definite Matrices

Suppose A is positive definite.

Show that

$$Ax = 0 \implies x = 0.$$

Proof skeleton:

Complete proof:

18.3 Concavity and Maximizers

Suppose $C \subseteq \mathbb{R}^n$ is convex and $f : C \rightarrow \mathbb{R}$ is concave.

Show that

$$\operatorname{argmax}_{x \in C} f(x)$$

is a convex set.

Proof skeleton:

Complete proof:

19 FINAL WORKSHOP: FROM IDEA TO PROOF

Consider the following result.

Proposition 1

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable and concave.

Suppose $x^* \in \mathbb{R}^n$ satisfies

$$\nabla f(x^*) = 0.$$

Then x^* is a global maximizer of f .

You may use the first-order characterization of concavity:

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) \quad \forall x, y \in \mathbb{R}^n.$$

Do not begin with the complete proof.

Step 1: State the goal precisely

To show that x^* is a global maximizer, what inequality must hold for every $y \in \mathbb{R}^n$?

Step 2: Identify the useful theorem

Apply the first-order characterization of concavity with

$$x = x^*.$$

What does this give?

Step 3: Use the first-order condition

Substitute

$$\nabla f(x^*) = 0.$$

What remains?

Step 4: Write a three-line proof skeleton

Step 5: Write the complete proof**20 FINAL CHECKLIST**

Before declaring a proof complete, ask:

1. Did I start from the assumptions?

Is it clear what objects I am considering and what I am assuming?

2. Did I state the important intermediate steps?

Or did I jump directly from the assumptions to the conclusion?

3. Did I justify the key transitions?

Which definition, assumption, theorem, or algebraic fact supports each important step?

4. Did I use the assumptions correctly?

Where exactly did each important assumption enter?

5. Did I actually reach the stated goal?

Is the final line exactly what the problem asked me to establish?

6. If the statement has two directions, did I prove both?**7. Could another reader reconstruct my reasoning?****21 TAKEAWAY**

The three stages we have practiced so far are:

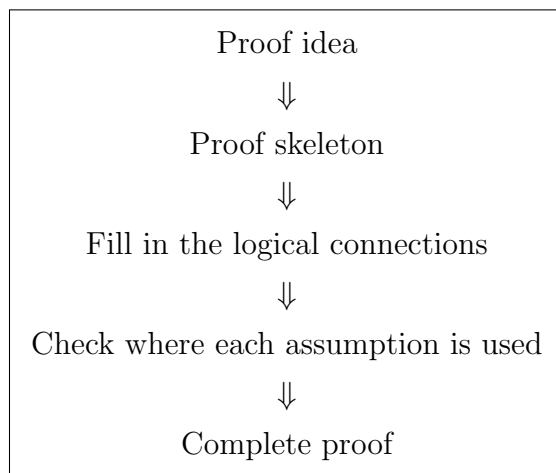
Deconstruct	→	Explore	→	Construct
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Deconstruction tells us what the problem means.

Exploration helps us discover a route.

Construction turns that route into a rigorous argument.

A useful workflow is:



A proof is complete when the reader does not have to guess why an important step follows from the previous one.